
Write-Time Governance for Persistent-State Integrity in AI Agents

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Abstract

1 Persistent agent memory is a reliability boundary: invalid writes can persist across
2 sessions and repeatedly affect later decisions, while answer-time filtering may
3 hide a conflict without repairing durable state. We present a write-time govern-
4 ance layer that treats memory updates as admissible state transitions; candidate
5 assertions undergo entity, schema, temporal, evidence, and contradiction checks
6 before commitment, and unresolved conflicts are quarantined with provenance
7 instead of entering active memory. On synthetic long-horizon tasks, read-time
8 filtering reduces contradictions from 14.49 to 1.45 per 100 responses but leaves
9 50.72 durable violations, whereas our governed system achieves a comparable con-
10 tradiction rate (1.26) while eliminating covered durable violations at a task-success
11 cost under ambiguous conflicts (77.29 to 59.03). The integrity gains generalize:
12 violations fall from 7.14 to 0.012 on MultiWOZ with 99.96 task success, from
13 106.38 to 0.00 on cached-oracle LongMemEval with 100 task success, and from
14 801.39 to 0.00 on raw LongMemEval while matching ungoverned task success
15 (54.17). In a controlled query-only stress test, governance blocks all 560 entity-
16 linking-evasive writes, whereas ungated baselines accept all 560. These results
17 establish persistent-state integrity under the evaluated constraints and threat model,
18 not universal security or general answer-quality superiority.

19 1 Introduction

20 Long-running AI agents retain state across sessions, making memory part of the operational substrate.
21 Persistent experience shapes downstream behavior Park et al. [2023], Shinn et al. [2023], Wang et al.
22 [2024], and memory-centric architectures make long-term memory a first-order systems issue Packer
23 et al. [2023], Zhong et al. [2024]. Yet practical stacks commonly store weakly structured summaries,
24 chunks, or embeddings optimized for recall rather than validity. Long-horizon benchmarks expose
25 failures in conversational recall, knowledge updating, evolving state, and sustained planning Maharana
26 et al. [2024], Wu et al. [2025], Zhao et al. [2026], Luo et al. [2025]. MemOps further shows that final-
27 answer accuracy can conceal faulty operation traces and that reconstructing ordered state trajectories
28 is substantially more fragile than binding individual targets Hao et al. [2026]. MemoryArena finds
29 that performance on recall-style memory benchmarks does not transfer reliably to interdependent
30 multi-session action, attributing part of the gap to memory optimized for generic recall, compression,
31 or similarity retrieval rather than task-relevant belief-state tracking He et al. [2026]. Once durable,
32 such errors are repeatedly retrieved and reused. Persistent memory is therefore a reliability boundary:
33 a transient error can become reusable state that affects many later decisions.

34 We treat each durable-memory update as a governed write. Before commitment, a candidate assertion
35 is checked for its type, entity, evidence, temporal validity, and conflicts with high-confidence memory.
36 Our governance layer represents entities, tasks, events, claims, documents, and decisions as typed
37 graph assertions with confidence, provenance, temporal, and lifecycle metadata. Candidates are

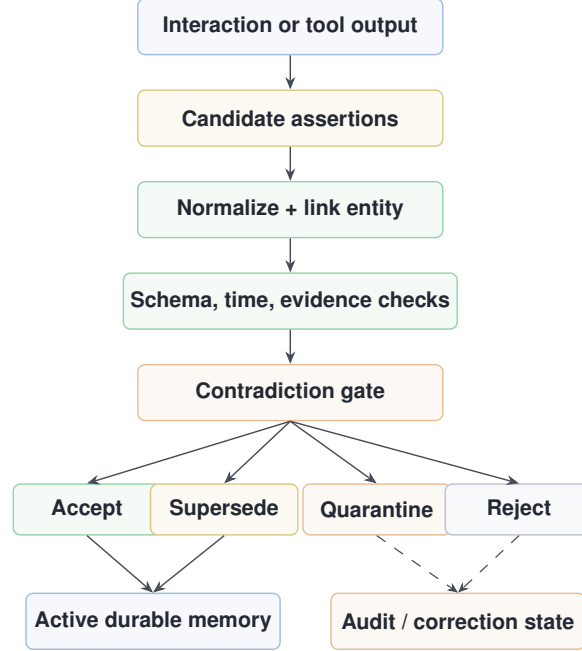


Figure 1: Write-time governance pipeline. Durable memory updates are treated as admissible state transitions: candidate assertions are linked, checked, and routed before they can become active reusable memory.

38 normalized, entity-resolved, and validated before entering active state Moreau and Missier [2013],
 39 World Wide Web Consortium [2017], Al-Moslmi et al. [2020]; unresolved conflicts are quarantined
 40 with traceable explanations. Figure 1 summarizes this control point.

41 Retrieval combines symbolic graph queries with semantic fallback and reliability-aware reranking.
 42 Unlike retrieval augmentation and query-time correction Lewis et al. [2020], Edge et al. [2024],
 43 Yan et al. [2024], governance constrains memory formation itself. We test this distinction on four
 44 synthetic long-horizon settings and with a read-time contradiction-filtering comparator (Figure 2).
 45 MultiWOZ oracle slots and cached-oracle and raw LongMemEval further test trusted updates and
 46 end-to-end extraction Wu et al. [2025].

47 Our contributions are: (1) an implemented typed write-time governance mechanism evaluated by
 48 direct inspection of committed memory state; (2) an assertion lifecycle spanning accepted, superseded,
 49 quarantined, and rejected states; (3) a cross-benchmark comparison of read-time suppression and
 50 write-time integrity, including controlled entity-linking evasion; and (4) a bounded-adjudication
 51 ceiling analysis. The last is diagnostic: label-informed review recovers 86% of the synthetic task-
 52 success cost, but learned and heuristic policies do not reproduce that ceiling. We evaluate persistent-
 53 state integrity under the implemented constraints and threat model, not universal security or general
 54 answer-quality superiority.

55 2 Related Work

56 2.1 Memory and Retrieval for LLM Agents

57 Persistent-agent memory research has largely optimized retrieval-time utility while assuming indexed
 58 state is sufficiently valid. ReAct, Generative Agents, Reflexion, and Voyager established reasoning-
 59 acting loops, persistent behavior, and lifelong skills Yao et al. [2023], Park et al. [2023], Shinn et al.
 60 [2023], Wang et al. [2024]; newer architectures address long-term memory management Packer et al.
 61 [2023], Zhong et al. [2024], Chhikara et al. [2025]. Towards LifeSpan Cognitive Systems identifies
 62 filtering, experience merging, and conflict resolution as requirements for continuously updated agents
 63 Wang et al. [2025]; it presents a system-level research agenda, whereas OCMR operationalizes this
 64 requirement as typed write-time admission with provenance, quarantine, and measurable durable-state

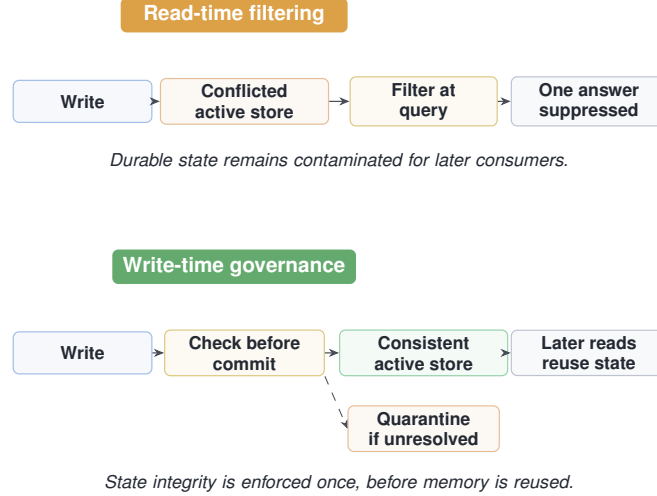


Figure 2: Control-point contrast. Read-time filtering can hide conflicts for a query, but the active store remains inconsistent; write-time governance prevents unresolved conflicts from becoming ordinary durable state.

constraints. Zep maintains temporally evolving facts in a knowledge graph Rasmussen et al. [2025], while LoCoMo and LongMemEval show that memory errors accumulate and degrade reliability Maharana et al. [2024], Wu et al. [2025]. MemoryArena couples memory to environment feedback and interdependent action; its results show that external memory and RAG do not consistently outperform verbatim long context and that structured 2D memory alone does not lead overall He et al. [2026]. STALE further isolates implicit conflicts in which later observations invalidate earlier memories without explicit negation Chao et al. [2026]. Retrieval- and graph-augmented generation improve factuality by conditioning on external evidence, including corrective and self-reflective variants Lewis et al. [2020], Asai et al. [2023], Edge et al. [2024], Gutiérrez et al. [2024], Yan et al. [2024], and proactive-memory work studies when stored state should intervene in an agent’s next action Wu et al. [2026]. These systems primarily decide what to retain, revise, or retrieve; our controlled comparison asks when a candidate assertion may cross the durable commit boundary.

MemOps is the closest evaluation prior: it represents remember, forget, update, reflect, and composed operations with gold targets, transitions, and evidence spans, then uses response accuracy and GPT-4o-judged operation diagnostics Hao et al. [2026]. OCMR does not claim operation-level diagnosis; it instead inspects accepted durable state directly, without an LLM judge, and treats quarantine as an explicit write outcome. EverMemOS organizes dialogue into provenance-bearing MemCells and MemScenes, adds validity-bounded foresight and recency-aware profile conflict tracking, and performs time-validity filtering during retrieval Hu et al. [2026]. These constructs overlap with our temporal and provenance metadata, but EverMemOS does not enforce typed relation signatures or cardinality admission and does not expose a quarantine state; its conflict-detection capability is evaluated qualitatively rather than by a durable-store integrity metric.

2.2 Structured and Symbolic Consistency

Knowledge graphs provide typed entities and relations for interpretable reasoning Liu et al. [2024]; entity linking Al-Moslmi et al. [2020], Wu et al. [2020] and validation languages World Wide Web Consortium [2017], Labra Gayo et al. [2018] specify constraints over admissible graph states. Truth-maintenance systems Doyle [1979], de Kleer [1986], dialogue consistency and factuality checking Chern et al. [2023], Min et al. [2023], Sato et al. [2024], and provenance-aware data management Moreau and Missier [2013] motivate the accepted, superseded, quarantined, and rejected lifecycle with source lineage. Temporal KG and dynamic graph reasoning Trivedi et al. [2017], Xu et al. [2022], Liu et al. [2022] motivate temporal sanity and interval-aware checks. Closest to our control point, TOKI formalizes contradiction resolution as bitemporal write-time operators with isolation and provenance contracts Wang [2026]. This concurrent system provides a stronger temporal formulation;

our evidence instead isolates typed admission versus read-time filtering and measures quarantine and entity-linking-evasion behavior.

2.3 Oversight and Alignment

Human oversight of automation depends on calibrated allocation of authority and trust, not nominal presence alone Parasuraman et al. [2000], Lee and See [2004]. In persistent agents this extends from answer alignment to state alignment: whether information available to future behavior remains admissible. Learning-to-defer work formalizes when a model should route a case to an expert Madras et al. [2018], Mozannar and Sontag [2020]; our bounded-review diagnostic asks whether quarantined writes are recoverable and whether write-time features support selective escalation. Poisoning agent memory or retrieval corpora can steer downstream behavior Chen et al. [2024], Zou et al. [2025]; MINJA demonstrates query-only memory injection Dong et al. [2025], and MPBench systematizes memory-write channels and structural vulnerabilities Dash et al. [2026]. MemPoison further shows that record-local write-time checks can miss compositional and dormant corruption Gao et al. [2026], which bounds our covered-state claim. We treat review as bounded adjudication of already-governed writes, not evidence that the governance layer itself recovers withheld utility.

2.4 Positioning and Novelty

The proposed layer occupies a control point between agent memory, retrieval-time correction, and symbolic consistency management. In MemoryArena’s 0D/1D/2D taxonomy, our governed system uses a 2D structured representation, but contributes an orthogonal admissibility layer for deciding whether proposed state may become durable He et al. [2026]. Agent-memory systems primarily decide what to retain and retrieve; RAG and corrective-RAG improve evidence access or repair an answer after content has entered an index; truth-maintenance systems revise belief sets; and learning-to-defer routes uncertain decisions to an expert. TOKI independently formalizes a closely related write-time contract. Our contribution is the empirical control-point formulation: typed, provenance-bearing state transitions are enforced before an assertion becomes reusable context, and write-time admission is compared directly with read-time contradiction filtering. The accepted, superseded, quarantined, and rejected lifecycle supports the measured integrity, review-cost, and controlled-evasion analyses. We do not claim a new ontology language, entity linker, retrieval model, operation-level memory diagnosis, or the first write-time memory protocol.

3 Method

3.1 Overview and Problem Formulation

We consider long-horizon agent operation where each interaction step t yields an input x_t (user text, tool output, or document evidence) and may update durable memory state $\mathcal{M}_t$. A candidate extractor maps x_t into zero or more assertion candidates $C_t = \{a_1, \dots, a_n\}$. Each candidate contains a subject mention, predicate, object value or object mention, evidence span, source identifier, observation time, optional validity interval, and extractor confidence. Existing memory systems often append semantically plausible content directly into storage, then rely on retrieval-time filtering. This permits logically inconsistent or stale facts to persist and later degrade reasoning quality.

We propose a governed memory architecture in which each write is treated as a state transition. Evidence enters a pipeline constrained by ontology rules; active assertions update the memory store and vector index; unresolved conflicts are routed to quarantine; and hybrid retrieval consults only active state unless conflict inspection is requested. A candidate write c_t is processed by a validation pipeline before affecting trusted memory:

$$\phi(c_t, \mathcal{M}_{t-1}) \in \{\text{accept}, \text{supersede}, \text{quarantine}, \text{reject}\}. \quad (1)$$

Only accepted and superseding writes enter active memory for downstream retrieval and answering. Quarantined writes remain available for audit and future correction, but are excluded from ordinary answer generation unless a user explicitly asks for unresolved conflicts.

Covered-state invariant. Let $\mathcal{C}$ denote the schema, cardinality, temporal, evidence, and lifecycle constraints implemented by the governance layer. Conditional on correct detection of those covered

constraints, every committed active state must satisfy $\mathcal{M}_t^+ \models \mathcal{C}$; a candidate that would violate this invariant is superseded, quarantined, or rejected rather than added as an ordinary active assertion. This is a guarantee only for constraints represented in $\mathcal{C}$, not for missing ontology rules or extraction errors outside the validator’s scope.

Notation. $c(a)$ denotes confidence of assertion a , $e(a)$ denotes the count of supporting evidence units, $A \perp B$ denotes contradiction under ontology constraints, τ is the contradiction-confidence threshold, Δ_c is the correction margin, and assertion status is in {accepted, superseded, quarantined, rejected}.

3.2 Threat Model and Scope

We consider both benign corruption and adversarial manipulation of persistent memory. An error source or attacker may supply user text, retrieved documents, or tool outputs that induce plausible but stale, contradictory, entity-confused, temporally invalid, or weakly evidenced candidate writes. The protected assets are the integrity of active durable state and the provenance trail used to audit state transitions. The governance layer aims to prevent a candidate that violates an implemented constraint in $\mathcal{C}$ from becoming ordinary reusable memory, and to contain unresolved conflicts in quarantine.

The guarantee is conditional on the validator, schema, trusted-source labels, and commit manager remaining uncompromised. An attacker who can alter those components, forge an authoritative source outside the provenance model, or exploit a constraint absent from $\mathcal{C}$ is out of scope. The layer also cannot prove the truth of a well-formed, non-conflicting false assertion or recover facts the extractor never proposes. Quarantine is therefore a containment mechanism, not a correctness proof. In addition to benign corruption, we evaluate a controlled query-only attacker that controls injected entity-mention surfaces but cannot read or modify the store, prompts, model weights, validator, or commit manager. This test covers specified entity-linking evasions; it is not unrestricted adaptive red-teaming, system-level compromise testing, or a universal security guarantee.

3.3 Memory Representation

The governed store uses a typed property graph $\mathcal{G} = (\mathcal{V}, \mathcal{E})$ with domain nodes and first-class assertion nodes, following the use of explicit graph structure for interpretable reasoning while adding write-time validation before graph mutation World Wide Web Consortium [2017], Liu et al. [2024]. The implemented ontology contains 12 entity types and 15 predicates, expanding to 68 admissible source–predicate–target signatures; Appendix A gives the complete schema and an abridged graph view. Domain nodes represent people, organizations, projects, tasks, events, claims, decisions, documents, assertions, slots, and controlled value nodes, while assertion nodes represent candidate facts as first-class objects rather than as unannotated edges:

$$a = \langle s, p, o, \kappa, c, e, \sigma, \pi, [t_s, t_e], v \rangle. \quad (2)$$

Here s, p, o are subject, predicate, and object; κ is the assertion type; c is extractor confidence; e is the evidence-unit count; σ is status; π is provenance; $[t_s, t_e]$ is the validity interval; and v is extractor/schema version. Provenance stores source URI or tool id, evidence span, observation timestamp, extractor version, and optional upstream derivation Moreau and Missier [2013]. Active memory $\mathcal{M}^+$ contains accepted assertions minus superseded assertions; quarantine memory $\mathcal{Q}$ stores unresolved or low-evidence conflicts. Only accepted assertions materialize as graph edges; superseded, quarantined, and rejected assertions remain auditable repository records but are excluded from the active graph.

3.4 Write-Time Governance and Contradiction Policy

Given input x_t , the pipeline executes extraction, normalization, entity resolution, schema checks, constraint checks, contradiction checks, and commit routing. Entity normalization maps mentions to canonical nodes using lexical aliases, dense entity-linking candidates, and ontology-compatible type constraints Al-Moslimi et al. [2020], Wu et al. [2020]. Schema validation checks domain/range compatibility, required attributes, cardinality, controlled vocabularies, and shape constraints. Temporal validation rejects malformed intervals, impossible orderings, and updates that violate known precedence constraints. Appendix A.2 enumerates the implemented W1–W8 stages and C1–C10 checks.

Algorithm 1 Deterministic write-decision policy

Require: Candidate assertion a , memory state $\mathcal{M}$

```
1: if schema/type validation fails then  
2:   return REJECT  
3: end if  
4: detect contradiction class and conflict set against accepted state  
5: if hard conflict and incumbent evidence dominates then  
6:   return QUARANTINE  
7: else if correction intent and  $c(a) - c(a_{old}) > \Delta_c$  and  $e(a) \geq e_{min}$  then  
8:   return SUPERSEDE  
9: else if ambiguity unresolved then  
10:  return QUARANTINE  
11: else  
12:  return ACCEPT  
13: end if
```

196 Contradiction detection uses four deterministic classes: functional conflicts over single-valued
197 predicates, temporal conflicts over invalid intervals or lifecycles, polarity conflicts between a claim
198 and its negation, and state-transition conflicts such as rejected $\rightarrow$ accepted without new evidence.
199 In the implemented relation schema, only ASSIGNED_TO, HAS_STATUS, and HAS_VALUE are
200 single-valued (Appendix A); other predicates do not create a functional conflict merely by linking
201 one subject to multiple objects. A functional conflict covers two active assignees for one task; a
202 temporal conflict covers an access state that remains valid after its recorded expiration. These classes
203 adapt truth-maintenance ideas of conflict sets and belief revision to constrained agent memory Doyle
204 [1979].

$$A \perp B \wedge c(A) > \tau \wedge c(B) > \tau \wedge B \in \text{accepted} \Rightarrow A \not\vdash \text{accept}. \quad (3)$$

205 Unless otherwise stated, $\tau = 0.80$ and $\Delta_c = 0.1$ in the Qwen run. The evidence threshold e_{min}
206 requires at least one supporting unit, counted from a provenance source reference plus accepted
207 EVIDENCE_FOR edges into the subject. A candidate is rejected for non-recoverable schema or
208 evidence failure, quarantined when ambiguity or contradiction is unresolved, and allowed to supersede
209 only with correction intent, sufficient evidence, and confidence exceeding the incumbent by at least
210 Δ_c . Correction intent comes from explicit correction language, newer authoritative source timestamps,
211 or tool outputs marked as replacements; otherwise stronger incumbent evidence dominates.

212 The policy additionally defines a bounded review tier between admission thresholds. Writes whose
213 scores fall in this interval route to adjudication rather than terminal quarantine. A released write is
214 committed through the same commit manager as an ordinarily accepted write; for a single-valued
215 relation, commitment retires any incumbent value before activating the release. Mandatory schema,
216 temporal, evidence, and other hard checks are never releasable. Thus review can reconsider ambiguous
217 conflicts but cannot waive the system’s mandatory admissibility conditions.

218 Table 3 illustrates the ordinary lifecycle: a first fact is accepted, an authoritative status change super-
219 sedes its incumbent, and an unsupported second assignee is quarantined. For temporal constraints,
220 “access expired on June 1” and “access remains valid on June 10” are quarantined unless a renewal
221 event provides provenance for the later state.

222 3.5 Hybrid Retrieval and Answer Policy

223 Retrieval combines symbolic graph queries and semantic fallback, then reliability-aware reranking.
224 Query intents are mapped to templates for fact lookup, temporal check, contradiction check, planning
225 state, and evidence audit. Symbolic retrieval returns assertions satisfying entity, predicate, status,
226 and interval constraints; semantic retrieval provides fallback recall over assertion text and evidence
227 spans Lewis et al. [2020], Edge et al. [2024], Yan et al. [2024]. Candidate assertions are merged by
228 entity-key, predicate, and validity interval before scoring:

$$s(i, q) = \alpha \text{sim}(i, q) + \beta \text{graphRel}(i, q) + \gamma \text{conf}(i) + \rho \text{prov}(i) + \eta \text{recency}(i) - \lambda \text{contrPenalty}(i). \quad (4)$$

Table 1: Baseline configurations.

Arm	Retrieval	Write-time governance	Purpose
B0 Text	Vector over serialized memory	Off	Text/vector anchor
B1 Ontology	Symbolic graph only	Schema only	Retrieval ablation
B2 Hybrid	Graph + semantic	Off	Closest ungoverned match
B_{RAG}	Vector text evidence only	Off	Vanilla RAG comparison
B_{RTCF}	Hybrid + read-time filter	No write gate	Read-time repair probe
B_{sup}	Hybrid	Latest-value overwrite only	Supersession-only ablation
B_{evi}	Hybrid	Evidence-focused write governance	Evidence-validity ablation
B_{memgpt}	LLM-managed memory	Insert/update/skip; no ontology gate	Raw LongMemEval comparator
B3 Governed	Hybrid + reliability metadata	Schema + gate + quarantine + provenance	Governed system

All terms are normalized to $[0, 1]$ on the validation split. The contradiction penalty suppresses unresolved or quarantined conflicts, and answers are generated from top- K active assertions with evidence ids, source timestamps, and conflict annotations when relevant.

4 Experimental Setup

4.1 Tasks and Evaluation Protocol

We evaluate memory reliability under four long-horizon settings: (i) *multi-session factual recall*, (ii) *contradiction-heavy update streams*, (iii) *temporal ordering and interval reasoning*, and (iv) *multi-step planning with evolving task state*, following stress patterns from long-horizon memory benchmarks Wu et al. [2025], Zhao et al. [2026]. Each scenario is a trajectory of interleaved user inputs and tool outputs, so writes and reads co-evolve.

All arms share extraction, normalization, entity resolution, commit, indexing, reranking, evidence packaging, traces, prompts, and checkpoints; only retrieval and governance switches differ. Table 1 isolates the architectural intervention and purpose of each comparison arm, while Table 2 identifies the distinct reliability property tested by each evaluation setting. B_{sup} isolates latest-value supersession, while B_{evi} isolates evidence-focused write governance; the cached-oracle LongMemEval artifact uses the same B_{evi} label for its benchmark-validated supersession comparator. The raw LongMemEval-only B_{memgpt} uses an LLM to choose insert, update, or skip without an ontology gate. Decisive metrics use four scenario classes with 25 trajectories each and five seeds per method. The class-stratified 60/20/20 split has no trajectory overlap; entity-resolution and evidence-required stresses are reported separately. Appendix C gives the six implementation categories, operational scoring rules, and disaggregated saved results.

The review experiment uses leave-one-seed-out evaluation. For each held-out seed, every arm is scored on all 156 examples and the escalation policy is fit only on the other seeds. The protocol nevertheless shares the generator’s vocabulary: 62.6% of held-out session text also occurs in the fitting pool (61.5–64.4% across seeds), concentrated at 94% in `evidence_required_decisions`. This bounds rather than eliminates parameter reuse. It also makes the selectivity null conservative, because the learned policy still fails to discriminate under conditions maximally favorable to vocabulary reuse. The reproduced contradiction rates match the published comparison exactly (B0 14.49 and B3 1.26). The published quarantine artifact contains 1,198 quarantines, including 835 benchmark-label-discordant cases; an independent rerun differs by one write, recording 1,199 and 836, respectively (rate 0.697). These checks license adding the leave-one-seed-out rows to Table 4 while keeping their same-run baselines explicit.

Model and extractor setup. Extraction uses local Qwen2.5-14B-Instruct (bf16); embeddings use all-MiniLM-L6-v2. Both are fixed across arms and documented in the released configurations. Model choice affects absolute recall, while the controlled comparison isolates governance effects.

Real-data validation setup. Beyond the synthetic stress benchmark, we evaluate MultiWOZ 2.2 validation dialogues using oracle slot extraction. Each dialogue-state slot is represented as a single-valued memory assertion, and changed slots are treated as authoritative updates, isolating cardinality governance from dialogue-state-tracking error.

Table 2: Role of each evaluation setting.

Setting	What it tests
Synthetic stress	Governed writes under controlled contradiction, temporal, and planning pressure.
Read-time probe	Whether answer-time filtering repairs contaminated durable state.
MultiWOZ slots	Single-valued governance under trusted dialogue-state updates.
LongMemEval oracle	Supersession of changing user facts independent of extraction error.
LongMemEval raw	End-to-end behavior under noisy extraction and entity linking.

Table 3: Representative write-time governance decisions.

Write scenario	Candidate assertion	Decision
New ownership fact	Rupert $\rightarrow$ OWNS $\rightarrow$ Draco	accept
Authoritative status update	T7 $\rightarrow$ HAS-STATUS $\rightarrow$ in progress	supersede
Second assignee; no correction	T15 $\rightarrow$ ASSIGNED-TO $\rightarrow$ Mallory	quarantine

We also evaluate LongMemEval’s knowledge-update subset Wu et al. [2025]. A cached-oracle arm uses 47 update examples and 30 abstention examples with Qwen-extracted, benchmark-validated value trajectories, isolating governed state transitions from extraction error. B0 and B2 accept all extracted fact states, while B_{evi} and B3 supersede stale assertions under authoritative updates.

An end-to-end LongMemEval arm runs the Qwen extractor over raw multi-session haystacks and caches SLOT-HAS-VALUE-SLOTVALUE candidates. This arm measures noisy extraction and entity resolution, while abstention examples test refusal when no grounded answer exists.

Together, the synthetic benchmark tests governed writes under controlled conflict pressure, the read-time comparator separates answer-time suppression from store repair, MultiWOZ and cached LongMemEval isolate trusted supersession, and raw LongMemEval diagnoses extraction and linking limits.

4.2 Metrics

We track three decisive metrics: contradiction rate per 100 responses ($\downarrow$), durable-write constraint violation rate ($\downarrow$), and long-horizon task success/plan completion ($\uparrow$), separating answer behavior from memory-operation and state-update reliability emphasized by recent benchmarks Wu et al. [2025], Zhao et al. [2026], Hao et al. [2026]. Task success is answer or plan-completion correctness only, implemented as mean answer-token recall against the gold answer or expected plan state; it is intentionally decoupled from conflict surfacing and is not blended with contradiction handling. Durable-write constraint violations measure contradictory durable state left accepted in memory: specifically, single-valued relations with two or more accepted objects, reported as a rate per 100 responses. Unlike MemOps’ response-level operation diagnostics Hao et al. [2026], this measure is computed directly from committed store contents without an LLM judge, while write outcomes separately expose whether a candidate was accepted, superseded, quarantined, or rejected. Reported diagnostics additionally include memory-induced hallucination rate, review demand, false-quarantine

Table 4: Primary decisive metrics. Primary rows report mean $\pm$ 95% Student- t CIs over five seed-level aggregates (25 trajectories per category per seed); bounded-review rows use five held-out-seed folds. Halluc. denotes hallucination rate; Rev./100 denotes reviews per 100 writes.

Arm	Task $\uparrow$	Halluc. $\downarrow$	Contrad. $\downarrow$	Viol. $\downarrow$	Rev./100 $\downarrow$
<i>Primary governance evaluation</i>					
B0 Text	77.29 $\pm$ 0.42	0.302	14.49 $\pm$ 0.00	50.72 $\pm$ 2.78	0
B1 Ontology	35.75 $\pm$ 2.85	—	14.49 $\pm$ 0.00	50.72 $\pm$ 2.78	0
B2 Hybrid	73.43 $\pm$ 1.12	0.632	14.49 $\pm$ 0.00	50.72 $\pm$ 2.78	0
B_{sup} supersession	87.83 $\pm$ 0.89	—	14.49 $\pm$ 0.00	64.54 $\pm$ 3.36	0
B_{evi} evidence	84.93 $\pm$ 1.43	—	14.49 $\pm$ 0.00	13.82 $\pm$ 0.68	0
B_{RAG} RAG	62.51 $\pm$ 0.91	—	14.49 $\pm$ 0.00	50.72 $\pm$ 2.78	0
B_{RTCF} control-point	72.66 $\pm$ 0.80	—	1.45 $\pm$ 0.00	50.72 $\pm$ 2.78	0
B3 Governed	59.03 $\pm$ 3.78	0.527	1.26 $\pm$ 0.33*	0.00 $\pm$ 0.00*	0
<i>Leave-one-seed-out recovery ceiling</i>					
$B3_R$ informed review	73.91 $\pm$ 1.74	0.526	1.26 $\pm$ 0.26	2.11 $\pm$ 0.67	59.8
$B3_R$ random, matched to informed	75.36 $\pm$ 1.77	0.544	3.48 $\pm$ 0.93	2.05 $\pm$ 0.19	58.9
$B3_R$ heuristic review	78.26 $\pm$ 1.49	0.529	12.75 $\pm$ 1.06	2.07 $\pm$ 0.53	58.2
$B3_R$ random, matched to heuristic	76.91 $\pm$ 2.35	0.538	7.54 $\pm$ 1.76	1.74 $\pm$ 0.29	58.4
$B3_R$ release-all	79.32 $\pm$ 1.65	0.520	14.49 $\pm$ 0.00	1.93 $\pm$ 0.51	57.8

audit counts, entity-resolution precision/recall, attack acceptance and quarantine, and scoped efficiency instrumentation.

Durable constraint violations are invalid active store states, such as multiple active values for one of the three single-valued predicates enumerated in Appendix A, normalized per 100 evaluated responses or examples. For the primary evaluation, each method–seed run is first aggregated over its 100 trajectories; reported CIs are Student- t intervals over the five seed-level aggregates. Primary significance tests compare paired seed-level aggregates for matched seeds ($n = 5$), not individual responses or trajectories, with Holm–Bonferroni correction across the three decisive metrics at $\alpha = 0.05$. The bounded-review rows use the five held-out-seed folds as the interval unit.

Significance. In Table 4, bold marks the best displayed value and * marks significant safety improvement after Holm–Bonferroni correction ($p < 0.05$) over paired seed-level aggregates. Against the strongest baseline for each metric, B3 reduces durable violations relative to B_{evi} (corrected $p = 1.81 \times 10^{-6}$; paired $d = -25.08$), while B_{sup} has higher raw task success than B3 (corrected $p = 7.18 \times 10^{-5}$; paired $d = -9.01$ for B3 minus B_{sup}). B3’s contradiction-rate difference from B_{RTCF} is not significant ($p = 0.178$; paired $d = -0.73$). Zero-width contradiction CIs reflect identical seed-level values under deterministic templates, not zero population uncertainty.

The high B_{sup} task score is not an integrity gain. Because its narrow HAS_VALUE rule is inactive, the arm retains incompatible active facts; hybrid retrieval can still surface the expected answer token from that inconsistent store. It therefore reaches 87.83 task success while leaving 64.54 durable violations and a 14.49 contradiction rate. B_{evi} lowers durable violations to 13.82 but leaves the contradiction rate at 14.49. B3 instead trades raw answer-token recall for governed admission, reducing those covered violations to zero.

4.3 Stress Benchmark Design

To stress persistent-memory brittleness, we construct controlled perturbation classes: (1) mutually conflicting updates, (2) entity alias ambiguity with partial context, (3) temporal overlap inconsistencies, and (4) confidence inflation under incomplete evidence, grounded in state-update, target-binding, and implicit-conflict failures studied in long-horizon memory benchmarks Zhao et al. [2026], Chao et al. [2026], Hao et al. [2026]. Each class is replayed across multi-session trajectories with increasing perturbation intensity.

For alias scenarios, we quantify merging and fragmentation with entity-resolution F1 and the false-merge rate. The Qwen run is conservative: precision is 1.00, recall is 0.185, F1 is 0.312, and the sampled false-merge rate is 0.00. Detailed category and intensity breakdowns appear in the released logs; the main text uses them only to locate the recall cost.

Table 5: Mechanism-critical ablations; five seeds and 25 trajectories per category.

Variant	Contrad. ↓	Viol. ↓	Task ↑
Full governed	1.26*	0.00*	59.03
w/o typed schema	1.26*	0.00*	59.03
w/o contradiction gate	14.49	50.72	73.43
w/o provenance scoring	1.26*	0.00*	59.03
w/o hybrid routing	1.26*	0.00*	63.77

325 4.4 Controlled Entity-Linking-Evasion Stress Test

326 We additionally test whether a query-only attacker can evade write-time contradiction checks by
 327 varying only the subject mention. The suite spans seven axes—novel aliases, spelling variants,
 328 spacing variants, partial identifiers, adaptive aliases, role descriptions, and unrelated aliases—with
 329 original and freshly mutated surfaces. Each original or mutated evasive suite contains 280 attempted
 330 adversarial writes: eight cases per axis under five seeds (56 trials per seed). We compare text memory
 331 (B0), ungoverned hybrid memory (B2), latest-value supersession (B_{sup}), full governance (B3), and
 332 B3 with the C7 fail-closed linkage-attribution check disabled. The attacker cannot inspect durable
 333 memory or alter system components; the oracle determines whether an injected surface denotes an
 334 already protected subject.

335 To measure collateral cost, a separate benign suite contains 320 writes spanning legitimate aliases,
 336 short and underscored identifiers, leading-zero variants, genuinely new entities, and intentionally
 337 ambiguous unattributed writes. We report erroneous benign decisions separately from review burden:
 338 a correctly quarantined ambiguous write is not labeled a false positive, but it does reduce immediate
 339 utility because review is required. The suite is seeded and templated, so raw counts characterize
 340 the tested construction rather than an unrestricted attack population. Appendix D reports the frozen
 341 per-axis and benign-family counts.

342 4.5 Ablation Plan

343 We ablate typed schema constraints, contradiction gating and quarantine, provenance scoring, and
 344 hybrid routing. These diagnostics are not exhaustive; Section 5.7 identifies which mechanisms move
 345 the decisive metrics. Table 5 is contradiction-dominated: nearly every violation without governance
 346 is a single-valued contradiction. The other components are either admission-equivalent to C9 or affect
 347 retrieval rather than admission. We therefore add a component-isolating workload (Appendix B) in
 348 which each write violates exactly one check and none is a single-valued contradiction.

349 4.6 Reproducibility Controls

350 The anonymous artifact bundle will include trajectory scripts, prompt templates, schema files,
 351 extractor/reranker configurations, seeds, split identifiers, candidate assertions, routing decisions,
 352 store snapshots, response text, and metric traces. Its landing-page map separates primary multi-seed
 353 results, bounded-review folds, real-data diagnostics, threshold and quarantine audits, and the frozen
 354 entity-linking-evasion suite. The tracked security artifact freezes the generator revision, five seeds,
 355 seven attack axes, original and mutated suites, baseline configurations, C7 ablation, and benign-suite
 356 aggregates needed to reproduce Section 5.2. The anonymous bundle URL must be inserted in the
 357 submission version; the author-identifying development repository is not an anonymous artifact.

Table 6: Controlled entity-linking-evasion results across five seeds.

Arm	Original accepted ↓	Mutated accepted ↓	Mutated quarantined ↑
B0 Text	280/280	280/280	0/280
B2 Hybrid	280/280	280/280	0/280
B_{sup} supersession	280/280	280/280	0/280
B3 Governed	0/280	0/280	280/280
B3, C7 off	—	280/280	0/280

5 Results and Analysis

5.1 Overall Outcome on Decisive Metrics

Table 4 shows the governed system’s main trade-off. It reduces contradiction rate from 14.49 to 1.26 and durable-write constraint violations from 50.72 to 0.00. Against the read-time control-point comparator, contradiction suppression is similar (1.26 vs. 1.45; $p = 0.178$), but durable violations remain unchanged for the comparator (50.72 vs. 0.00 for B3), showing that answer-time filtering does not repair contaminated memory state. B0 retains higher raw task success (77.29 vs. 59.03), so the result is not Pareto dominance: B3 is a memory-state integrity layer for settings where contaminated durable state is costlier than abstention or conflict surfacing. The bounded-review ceiling in Section 5.3 shows that 86% of this recall cost is recoverable in principle while retaining 100% of the contradiction-rate improvement; it does not establish that the governed system or a deployable reviewer achieves that recovery.

5.2 Containment Under Entity-Linking Evasion

Table 6 reports a sharp control-point separation. B0, B2, and B_{sup} accept every evasive write in both the original and mutated suites (280/280 per suite), whereas B3 accepts none (0/280 per suite) and quarantines all 280 mutated attempts. Disabling C7 while leaving the rest of B3 unchanged restores 280/280 mutated-suite acceptance, isolating fail-closed linkage attribution as the decisive containment mechanism on this workload. There is no observed between-seed variation in these rates; because the examples share seeded templates, we report counts rather than interpreting the zero variance as a population-level confidence interval.

On the 320-write benign suite, B3 makes 0/320 erroneous decisions under the suite labels. It nevertheless routes 40/320 intentionally ambiguous writes to review, for a 12.5% quarantine burden and 87.5% immediate utility success. Thus the security result is containment rather than cost-free resolution: fail-closed admission prevents the tested evasions from entering active state by converting unresolved attribution into explicit review work.

5.3 Bounded Review as a Recovery Ceiling

On the leave-one-seed-out runs, governance costs 17.49 task-success points (B0 76.33 versus B3 58.84) and buys 13.24 contradiction-rate points. Label-informed bounded review reaches 73.91 task success at contradiction rate 1.26, recovering 86% of the task-success cost while retaining 100% of

Table 7: Reviewer policies for bounded review.

Policy	Information and role
Informed	Conflict labels; non-deployable ceiling.
Heuristic	Cross-context identifier collision.
Random at rate p	Release-rate-matched no-skill control.
Release-/uphold-all	Release/retain every eligible quarantine.

387 the contradiction-rate gain. This is a ceiling: the informed reviewer reads `expected_conflict`, a
388 benchmark label unavailable in deployment.

389 Task success generally rises with admitted volume, so release-all is not the relevant attribution test. At
390 the informed reviewer’s release fraction, rate-matched random review retains 83% of the contradiction
391 gain whereas informed review retains 100%, an observed 17-point difference. Interpolation across
392 the release-rate frontier gives a similar descriptive difference of 16 points. The frozen artifact does
393 not retain the paired fold-level outputs needed for a valid inferential comparison, so we do not attach
394 a p -value or significance claim to these two differences. They are volume controls consistent with,
395 but not proof of, an adjudication-quality explanation.

396 The learned escalation result is negative. Across all nine reviewers, the learned escalation arm $B3_R$
397 and the full-quarantine routing arm $B3_Q$ agree on every metric, including review rate, because the
398 learned policy routes exactly the complete quarantine set. Constraint-failure features have zero
399 separability for benchmark-label-discordant quarantine: lift is $+0.0000$ on every seed and precision
400 equals the base rate over 1,199 quarantines. The write-time signal needed for selective routing is
401 absent even under the vocabulary-sharing protocol favorable to the learner.

402 A deployable cross-context identifier-collision heuristic also underperforms its rate-matched random
403 control. It releases 91.8% of held writes yet retains only 13% of the contradiction gain; random
404 release at 88.3% retains 53%, an observed 39-point deficit. As above, this is a descriptive comparison
405 because the paired fold-level outputs are not frozen for inference. Accordingly, the results bound
406 achievable adjudication at 86% recovery but do not supply a deployable method: neither the learned
407 router nor the tested heuristic approaches the label-informed ceiling.

408 5.4 Real-Data Validation on MultiWOZ

409 Table 8 reports MultiWOZ 2.2 with oracle dialogue-state slots. Because changed slots are authori-
410 tative, B3 supersedes stale values rather than quarantining them, reducing durable slot-cardinality
411 violations from 7.14 to 0.012 while preserving 99.96 task success. This tests trusted single-valued
412 updates, not end-to-end dialogue-state tracking.

413 5.5 Real-Data Validation on LongMemEval

414 The cached-oracle LongMemEval setting extends structured slots to open-domain user facts while
415 isolating supersession from extraction error. B0 and B2 leave stale and current values active, yielding
416 106.38 violations; B_{evi} and B3 preserve 100.0 task success, supersede 250 stale states, and reach 0.00
417 violations. In raw-text LongMemEval, B3 matches ungoverned task success (54.17) while eliminating
418 801.39 durable violations. The MemGPT-style arm reduces violations to 4.17 but does not eliminate
419 occasional incorrect inserts. Oracle performance of 100.0 confirms that raw end-to-end recall is
420 bounded by extraction and linking, not the gate. Abstention accuracy is 100.0 and false-answer
421 rate is 0.0 for every LongMemEval arm. EverMemOS reports 83.00 overall accuracy and 89.74 on
422 knowledge update over all 500 LongMemEval-S questions using GPT-4.1-mini and a three-judge
423 answer-scoring protocol Hu et al. [2026]; those values are not directly comparable to our Qwen-based
424 token-recall task success on 72 raw update examples, separate 30-example abstention evaluation, and
425 direct store-state audit.

426 5.6 Task-Wise Behavior Across Long-Horizon Scenarios

427 The category breakdown clarifies the failure mode: the recall cost is not uniform. B0, B2, and B_{RTCF}
428 retain stronger raw task success on factual QA, planning, and temporal reasoning, while B3 preserves
429 the safety properties reported in Table 4. The evidence-category loss is a retrieval failure rather
430 than a governance cost: ungoverned B2 already scores 26.15 against B0’s 63.85, so attributing it
431 to conservative quarantine is unsupported. Temporal performance is partly recoverable: B0 scores
432 85.38, B3 falls to 57.69, and bounded review reaches 67.69, recovering roughly 35% of that gap.
433 Evidence retrieval is the only category for which review shows no recovery.

434 5.7 Ablation Interpretation

435 Table 5 identifies contradiction gating as the dominant measured safety lever: removing it raises
436 contradiction rate from 1.26 to 14.49 and durable violations from 0.00 to 50.72. Read-time filtering

Table 8: Real-data validation: MW (1,000 dialogues), LM-O (47 updates; 30 abstentions), and LM-R (72 updates; 30 abstentions). A/S/Q/(R) denotes accept/supersede/quarantine/(reject).

Data	Arm	Task $\uparrow$	Contr. $\downarrow$	Viol. $\downarrow$	A/S/Q/(R)
MW	B0 Text	99.87	0.00	7.14	43550/0/0
	B2 Hybrid	99.87	0.00	7.14	43550/0/0
	B3 Governed	99.96	0.00	0.00	40540/3010/0
LM-O	B0 Text	100.0	0.00	106.38	485/0/0
	B2 Hybrid	100.0	0.00	106.38	485/0/0
	B_{evi}	100.0	0.00	0.00	235/250/0
	B3 Governed	100.0	0.00	0.00	235/250/0
LM-R	B0 Text	54.17	—	801.39	17697/0/0/0
	B2 Hybrid	54.17	—	801.39	17697/0/0/0
	B_{memgpt}	52.78	—	4.17	17115/544/0/0
	B3 Governed	54.17	—	0.00	17113/584/0/0

can hide contradictory answers but leaves that contaminated state intact. The other ablations are unchanged for identifiable reasons: graph domain/range checks absorb typed-schema removal, provenance affects reranking rather than admission, and removing hybrid routing slightly raises task success on this structural workload. We therefore treat the gate as decisive here, while schema, provenance, and hybrid routing support auditability and constraint classes isolated in Appendix B.

5.8 Operational Diagnostics

Threshold sweeps expose a safety–quarantine trade-off. Although an objective omitting durable violations favors $\tau = 0.95$, that setting restores 50.72 violations. We use $\tau = 0.80$, which keeps contradiction leakage near the floor and covered violations at 0.00; this is a workload-specific, not universal, choice. Entity resolution is deliberately conservative (precision 1.00, recall 0.185, F1 0.312, sampled false-merge rate 0.00), avoiding observed false merges while fragmenting high-alias updates and increasing quarantine. In offline replay, isolating each example’s store reduces label-valid quarantines from 297 to 0, implicating cross-example identifier reuse. Better alias resolution and source normalization remain priorities; the tested learned and heuristic reviewers do not make release deployable. No stable systems aggregate is frozen, so we make no cost claim.

6 Limitations

The synthetic benchmark is controlled; MultiWOZ and oracle LongMemEval isolate scoped update properties; raw LongMemEval remains a cached single-run diagnostic; and the appendix stress test uses engineered violations. LongMemEval does not close an agent–environment feedback loop or test interdependent subtasks, a gap made explicit by MemoryArena; evaluating governed writes inside that loop is an important next step He et al. [2026]. The entity-linking security suite is also seeded and templated: it covers a specified query-only attacker and seven mention-surface axes, not unrestricted adaptive red-teaming, compromised system components, or unseen attack families. Its 0/560 observed B3 acceptance therefore establishes containment on the tested construction rather than a universal failure probability. The benign suite records no labeled false-positive decisions, but 12.5% of writes require review; representative application traffic may have a different burden. Ontology completeness, provenance, and update-intent calibration bound governance quality. Review adds four limitations: informed review reads benchmark labels; every release breaks zero violations (0.00 to 2.11); review costs roughly 58 decisions per 100 writes and learned routing does not reduce it; and a decision rule simulates the adjudicator. Thus the study bounds achievable adjudication, not human behavior.

7 Conclusion

Persistent memory is a reliability boundary. Read-time filtering lowers visible contradictions to 1.45 but leaves 50.72 durable violations; write-time governance reaches 1.26 and 0.00. Real-data updates preserve task success when semantics are explicit. Under the controlled query-only security stress test, governance contains all 560 tested entity-linking evasions while the ungated baselines accept all 560, with a 12.5% benign-suite review burden. Together these results support covered persistent-state integrity under benchmark corruption and the specified adversarial stress model, not universal security or answer-quality superiority; label-informed recovery remains a non-deployable ceiling.

Impact Statement

Write-time governance can improve traceability by preventing covered invalid states from becoming reusable context, but incomplete schemas may suppress legitimate updates or encode bias, and poor calibration increases review burden. Validators cannot protect absent constraints or unseen attacks. High-impact use therefore requires subgroup monitoring, correction logs, explicit coverage documentation, and human override.

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A Implemented Typed Ontology

A.1 Graph Schema

The implementation represents the ontology as a directed multigraph whose nodes are typed entities and whose edges are accepted assertions keyed by predicate. Figure 3 shows the compact topology. It omits the two widest fan-outs, ABOUT and POSSIBLY_SAME_AS, to keep the diagram legible; Table 9 gives all 15 predicates and all 68 fully expanded type signatures. Superseded, quarantined, and rejected assertions remain as repository rows for provenance and audit, but never become graph edges.

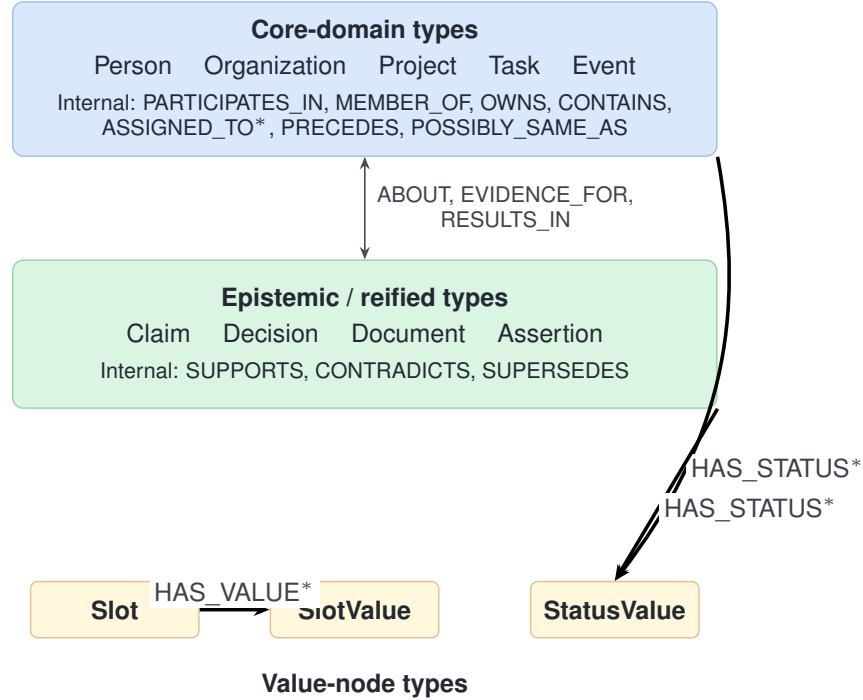


Figure 3: Typed-ontology topology grouped by node family. Thick arrows show single-valued edges to value nodes, and * marks every single-valued predicate, including ASSIGNED_TO in the core-family loop. The bidirectional family-level arrow summarizes predicates whose concrete signatures run between the core and epistemic families; Table 9 specifies every direction and source/target type exactly.

A.2 Implemented Constraints and Write Stages

The validator applies the ten checks in Table 10. “Reject” denotes a non-recoverable structural failure; “quarantine” preserves a recoverable but inadmissible candidate for review. C7 delegates conflict detection to W7 rather than duplicating it. A correction supersedes only when it exceeds the incumbent confidence by Δ_c and meets the evidence floor; an explicitly trusted authoritative update may supersede directly.

The write path is an eight-stage admission transaction:

- W1 **Extract**: convert source text into candidate facts.
- W2 **Normalize**: canonicalize values and mention surfaces.
- W3 **Resolve**: link mentions to existing entities.
- W4 **Build**: construct typed candidate assertions.
- W5 **Schema validate**: check structural ontology compatibility.
- W6 **Constraint validate**: apply C1–C10 against graph state.

Table 9: Complete implemented relation signatures. “Pairs” counts fully expanded source–predicate–target type triples. Only the three predicates marked single-valued can produce the functional conflicts counted by the durable-violation metric.

Predicate	Cardinality	Source type(s)	Target type(s)	Pairs
PARTICIPATES_IN	m:n	Person	Event	1
MEMBER_OF	m:n	Person	Organization	1
OWNS	m:n	Organization, Person	Project	2
CONTAINS	1:n	Project	Task	1
ASSIGNED_To*	m:1	Task	Person	1
PRECEDES	m:n	Event	Event	1
SUPPORTS	m:n	Claim	Claim, Decision	2
CONTRADICTS	m:n	Assertion, Claim	Assertion, Claim	4
EVIDENCE_FOR	m:n	Document, Event	Assertion, Claim, Decision	6
RESULTS_IN	m:n	Decision, Event	Event, Project, Task	6
ABOUT	m:n	Claim, Document	Decision, Event, Person, Project, Task	10
POSSIBLY_SAME_AS	m:n	Event, Organization, Person, Project, Task	Event, Organization, Person, Project, Task	25
SUPERSEDES	m:n	Assertion	Assertion	1
HAS_STATUS*	m:1	Claim, Decision, Organization, Person, Project, Task	StatusValue	6
HAS_VALUE*	m:1	Slot	SlotValue	1
Total fully expanded signatures				68

*Single-valued predicate: a second distinct object for the same subject is a functional contradiction.

Table 10: Complete implemented constraint catalogue. Actions are the default routing recommendations when a check fails.

ID	Check	Failure condition	Action
C1	Identity uniqueness	An identifier already belongs to a different entity type.	Reject
C2	Temporal sanity	An Event has an end timestamp earlier than its start timestamp.	Reject
C3	Acyclic PRECEDES	Adding the candidate edge would close a cycle in accepted PRECEDES edges.	Reject
C4	Completion evidence	A Task marked done lacks an incoming accepted Event–RESULTS_IN edge.	Quarantine
C5	Active assignee	An ASSIGNED_To target resolves to an inactive Person.	Quarantine
C6	Confidence bounds	Confidence is nonnumeric or outside [0, 1].	Reject
C7	Contradiction gate	A high-confidence candidate conflicts with a high-confidence accepted assertion and does not satisfy supersession conditions.	Quarantine [†]
C8	Decision evidence floor	A final Decision lacks the required accepted EVIDENCE_FOR edge from a Document or Event.	Quarantine
C9	Graph domain/range	The predicate is unknown, an endpoint type is unresolved, or resolved endpoint types violate the relation signature.	Reject
C10	Task-status transition	A non-correction Task transition is absent from the allowed transition map or uses an unknown status.	Quarantine

[†]A qualifying correction or trusted authoritative update is routed to supersede instead.

- 656 **W7 Contradiction check:** compare with accepted high-confidence knowledge.
- 657 **W8 Commit:** accept, supersede, quarantine, or reject; only accepted assertions are embedded
- 658 and materialized as graph edges.

659 B Component-Isolating Stress Diagnostic

660 Table 11 isolates the write-time checks that Table 5 cannot separate. The interpretation in Section 5.7

661 shows why: on the contradiction-dominated workload, removing typed schema (W5) is absorbed

662 by the graph domain/range check (C9, W6), and provenance/hybrid ablations act on scoring and

663 retrieval rather than on which assertions are admitted. We therefore construct a synthetic workload

664 of durable writes that are, by construction, *not* single-valued contradictions—so the contradiction

665 gate (C7) has no conflicting accepted assertion to fire on—yet each violates exactly one other

666 check: schema/domain-range (C9), temporal sanity (C2, an event whose end precedes its start),

667 the decision-evidence floor (C8, a final decision with no supporting evidence), and task-status

668 transitions (C4/C10, an unsupported completion and an illegal done→todo flip). The workload

669 mixes these poison writes with benign valid writes to measure precision.

670 We run the same inputs through four configurations built only from the existing governance toggles

671 (W5 = schema validation, W6 = constraint validation containing C1–C10, C7 = contradiction gate)

672 and count the invalid assertions left in the durable active store, by type. Extraction is a deterministic

673 offline oracle and storage is in-memory, so a single seed is sufficient, consistent with the other

674 deterministic tables.

Table 12: Complete implementation-category catalogue for the synthetic benchmark.

Implementation category	Short label	Controlled probe
longitudinal_factual_qa	Recall	Facts established over turns and recalled later.
multi_step_planning_entity_consistency	Plan	Owner and assignee consistency across sessions.
contradiction_heavy_update_stream	Conflict	Conflicting status updates to the same task.
temporal_reasoning_ordered_events	Temporal	Ordered events and PRECEDES reasoning.
entity_resolution_ambiguity	Entity	Alias surfaces referring to the same or different entities.
evidence_required_decisions	Evidence	Final decisions requiring supporting evidence under C8.

Table 11: Invalid active assertions in the component-isolation diagnostic (lower is better).

Arm ($W5/W6/C7$)	Schema	Evid.	Temp.	Status	Total↓
Ungoverned (off/off/off)	4	2	4	2	12
Gate-only (off/off/on)	4	2	4	2	12
Schema+Constraints (on/on/off)	0	0	0	0	0
Full governed (on/on/on)	0	0	0	0	0

C Extended Evaluation Protocol and Results

C.1 Synthetic Benchmark Construction and Scoring

The synthetic benchmark is seeded and generated offline. For each seed, it contains 25 generated examples in each of the six implementation categories in Table 12, plus six hand-authored anchors, for 156 examples. The anchors exercise a done-versus-not-started conflict, longitudinal recall, a project-owner conflict, inactive assignment (C5), a final decision without evidence (C8), and a PRECEDES cycle (C3). The saved Qwen evaluation uses seeds 1337, 7, 42, 99, and 2024. Each question stores canonical answer tokens, a Boolean expected-conflict label, and, where applicable, expected supporting-memory identifiers. Entity-resolution examples additionally store gold mention clusters.

Task success is 100 times the mean fraction of expected tokens found in the rendered answer and retrieved evidence. Contradiction rate is the number of expected conflicts not surfaced per 100 responses. Durable violations are single-valued subject–predicate keys left with at least two distinct accepted objects, normalized per 100 responses. Results are first aggregated within seed; confidence intervals and paired tests in the main paper use the five seed aggregates.

C.2 Category and Seed Disaggregation

Table 13 reports category-level task success from the saved Qwen result. It makes two nonuniform effects visible: B3 retains 100 on the entity category but falls most on recall, planning, and temporal questions; the evidence score of 26.15 is shared by B2, B_{sup} , B_{evi} , B3, B_{RAG} , and B_{RTCF} , so that loss is not specific to governance.

The saved stress summary also stratifies task success by perturbation intensity (Table 14). The decline from low to high intensity occurs for every arm; B3 moves from 80.00 to 63.33, while the ungoverned B0 anchor moves from 87.50 to 70.83.

Table 13: Task success (%) by implementation category, averaged over five seeds. Category labels are defined in Table 12.

Arm	Recall	Plan	Conflict	Temporal	Entity	Evidence
B0	98.08	86.67	3.70	85.38	100.00	63.85
B1	69.62	64.71	0.74	0.00	16.00	2.31
B2	98.08	90.20	3.70	85.38	100.00	26.15
B_{sup}	100.00	95.69	100.00	85.38	100.00	26.15
B_{evi}	100.00	100.00	100.00	53.85	100.00	26.15
B3	78.08	66.27	3.70	57.69	100.00	26.15
B_{RAG}	98.08	45.88	3.70	85.38	100.00	26.15
B_{RTCF}	98.08	87.06	3.70	85.38	100.00	26.15

Table 15: Saved seed-level aggregates in seed order 1337/7/42/99/2024. TS is task success, CR contradiction rate, and DV durable violations; all are rates per 100.

Arm	TS	CR	DV
B0	76.81/77.29/77.78/77.29/77.29	14.49/14.49/14.49/14.49/14.49	49.28/52.66/48.79/53.62/49.28
B1	32.85/38.16/33.82/37.20/36.71	14.49/14.49/14.49/14.49/14.49	49.28/52.66/48.79/53.62/49.28
B2	72.46/74.40/73.91/72.46/73.91	14.49/14.49/14.49/14.49/14.49	49.28/52.66/48.79/53.62/49.28
B_{sup}	87.44/87.92/88.89/86.96/87.92	14.49/14.49/14.49/14.49/14.49	63.29/66.67/61.84/68.12/62.80
B_{evi}	83.57/85.02/85.51/86.47/84.06	14.49/14.49/14.49/14.49/14.49	14.01/14.01/13.04/14.49/13.53
B3	54.11/59.42/58.94/62.32/60.39	1.45/1.45/1.45/0.97/0.97	0.00/0.00/0.00/0.00/0.00
B_{RAG}	61.35/62.80/62.80/63.29/62.32	14.49/14.49/14.49/14.49/14.49	49.28/52.66/48.79/53.62/49.28
B_{RTCF}	71.98/73.43/72.95/71.98/72.95	1.45/1.45/1.45/1.45/1.45	49.28/52.66/48.79/53.62/49.28

Table 14: Task success (%) by perturbation intensity.

Arm	Low	Medium	High
B0	87.50	89.17	70.83
B1	20.00	9.17	8.33
B2	87.50	90.00	71.67
B_{sup}	81.67	89.17	70.83
B_{evi}	72.50	75.00	62.50
B3	80.00	80.83	63.33
B_{RAG}	87.50	89.17	70.83
B_{RTCF}	87.50	89.17	70.83

698 C.3 Write Routing and Paired Tests

699 Table 16 exposes the durable routing decisions behind the aggregate metrics. The total is over all five
700 saved seed runs. B3 accepts 645 of 1,874 candidates, supersedes 22, and quarantines 1,207; B_{evi}
701 instead rejects 353 and supersedes 293 of 1,812 candidates. These arms therefore reach their integrity
702 results through materially different state transitions.

Table 16: Synthetic write outcomes aggregated over five seeds.

Arm	Cand.	Accept	Super.	Quar.	Reject
B0	1874	1296	22	556	0
B1	1874	1296	22	556	0
B2	1874	1296	22	556	0
B_{sup}	1812	1812	0	0	0
B_{evi}	1812	1166	293	0	353
B3	1874	645	22	1207	0
B_{RAG}	1874	1296	22	556	0
B_{RTCF}	1874	1296	22	556	0

Table 17: Paired seed-level tests for B3 against the strongest baseline for each decisive metric. Corrected p uses Holm–Bonferroni over the three tests.

Metric	Comparator	t	Corrected p	Paired d	Reject?
Task success	B_{sup}	−20.14	7.18×10^{-5}	−9.01	Yes
Contradiction	B_{RTCF}	−1.63	0.178	−0.73	No
Durable violations	B_{evi}	−56.09	1.81×10^{-6}	−25.08	Yes

703 C.4 Recorded Threshold Sweep

704 The saved sweep evaluates $J = \text{CR} + 0.5 \text{FQ} + 10 \text{ECE}$ on seed 1337, where FQ is false-quarantine
705 rate. As Table 18 shows, this scalar objective selects $\tau = 0.95$. The paper uses $\tau = 0.80$ because the
706 operating decision also includes the separate durable-state audit discussed in Section 5.7. The sweep
707 record itself does not serialize durable violations, so its selected threshold should not be read as an
708 integrity-optimal operating point.

Table 18: Saved B3 threshold sweep (seed 1337). CR and FQ are percentages; lower is better.

τ	CR	FQ	ECE	J
0.60	1.45	76.27	0.140	40.99
0.70	1.45	76.27	0.140	40.99
0.80	1.45	76.27	0.140	40.99
0.90	1.45	75.71	0.235	41.65
0.95	1.45	39.55	0.232	23.55

709 D Entity-Linking-Evasion Breakdowns

710 The frozen security artifact contains eight attempted writes per axis and seed. Table 19 aggregates the
711 five seeds. Every B3 attempt is quarantined for both original and freshly mutated mention surfaces,
712 including the adaptive families; no adversarial candidate becomes durable active state. These are
713 deterministic counts on the specified templated construction, not estimates for an open-world attack
714 population.

715 The same table disaggregates the 320-write benign suite. All 40 intentionally unattributed ambiguous
716 writes require review and therefore yield no immediate utility; the other seven families are admitted
717 without review. No family records a labeled false-positive decision.

Table 19: Frozen B3 entity-linking-evasion and benign-suite breakdowns across five seeds. A/Q denotes accepted/quarantined; benign utility denotes accepted or superseded writes.

(a) Evasion outcomes by attack axis

Axis	Attempts	Original A/Q	Mutated A/Q
Novel alias	40	0/40	0/40
Spelling variant	40	0/40	0/40
Spacing variant	40	0/40	0/40
Partial identifier	40	0/40	0/40
Adaptive alias	40	0/40	0/40
Role description	40	0/40	0/40
Unrelated alias	40	0/40	0/40
Total per suite	280	0/280	0/280

(b) Benign outcomes by family

Family	n	Errors	Review	Utility
Ambiguous unattributed	40	0	40	0
Genuinely new entity	40	0	0	40
Leading zero	40	0	0	40
Leading zero, attributed	40	0	0	40
Legitimate alias	40	0	0	40
Short identifier	40	0	0	40
Underscore	40	0	0	40
Underscore, attributed	40	0	0	40
Total	320	0	40	280

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1050 Justification: The Experimental Setup identifies Qwen2.5-14B-Instruct as the fixed extractor
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